

Automation of Vital Signs Monitoring Through Smart Sensors and GSM Communication for Elderly Care

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Abstract—The automation of vital-signs monitoring through smart sensors and GSM communication is presented as a low-cost technological proposal for strengthening real-time biomedical supervision, with a specific focus on older adults. The system integrates Internet of Things (IoT) technologies, biomedical sensors, a microcontroller, and wireless communication to continuously acquire, process, and transmit heart rate, blood-oxygen saturation (SpO_2), and body temperature. An on-device decision rule generates automatic alerts when physiological thresholds are exceeded; each alert, delivered as an SMS through the GSM network, details the critical parameters detected and advises the user to seek immediate medical attention, thereby supporting preventive supervision and a rapid response in risk situations. The prototype was built with an ESP32 microcontroller, a MAX30102 pulse-oximetry sensor, an MLX90614 non-contact infrared thermometer, and a SIM800L GSM module, complemented by a local web dashboard for real-time visualization. Beyond the implementation, this paper situates the design within the literature on remote health monitoring for the aging population, discusses the operating principles of the selected sensors, and analyzes the security, reliability, and validation challenges that condition the clinical adoption of such systems. The result is an accessible, replicable architecture that lowers the entry barrier to preventive remote monitoring in resource-constrained domestic and community settings.

Keywords—Internet of Things (IoT), vital-signs monitoring, biomedical sensors, GSM communication, digital health, elderly care, ESP32, photoplethysmography, remote patient monitoring

I. INTRODUCTION

Population aging is one of the defining demographic trends of the twenty-first century. According to the World Health Organization, the share of the world's population aged 65 years or older is projected to rise from about 10% in 2022 to 16% by 2050 [1]. Older adults are more susceptible to chronic and cardiovascular conditions, and to acute physiological deterioration that benefits from early detection [2]. This demographic pressure increases the demand for continuous health supervision while straining the human and financial resources of conventional care models [3].

Traditionally, vital-signs control depends on manual supervision or on equipment that requires the constant

presence of medical staff, which limits follow-up capacity and increases the risk of undetected health incidents among older adults living independently [4]. In response, systems based on the Internet of Things (IoT), biomedical sensors, and wireless communication make it possible to collect, process, and transmit physiological information in real time, automating supervision and supporting preventive decisions within the framework of ambient assisted living [5], [6].

This work proposes an intelligent monitoring system focused on older adults, integrating biomedical sensors, an ESP32 microcontroller, and GSM communication to issue automatic alerts that detail the detected parameters and recommend visiting the nearest medical center when irregularities occur. Unlike approaches that depend on permanent internet connectivity or specialized medical infrastructure, the proposal adopts a low-cost, GSM-based design that remains operational where Wi-Fi coverage is unreliable, an important consideration in domestic and rural settings. The contribution is twofold: a functional, replicable prototype, and a critical discussion—grounded in the recent literature—of the sensing principles, security, and validation requirements that govern the responsible deployment of such systems.

The remainder of the paper is organized as follows. Section II reviews the technological background and related work. Section III describes the system architecture, components, and implementation. Section IV presents a comparative analysis and discusses the results, including security and limitation considerations. Section V analyzes the projected impact on older adults, and Section VI concludes the paper.

II. TECHNOLOGICAL BACKGROUND AND RELATED WORK

A. IoT and Remote Health Monitoring

The Internet of Things enables the interconnection of electronic devices through communication networks capable of collecting, processing, and transmitting information in real time, and it has become a central paradigm in the automation of medical supervision [5]. In the healthcare domain, IoT has been applied to rehabilitation follow-up, daily-activity

monitoring, medication reminders, and the remote monitoring of patients' vital signs [7]. Systematic reviews of wearable and IoT solutions for elderly care report consistent benefits in objective, reliable, and remote monitoring through ambient assisted living, while also identifying recurring challenges such as sensor accuracy, battery autonomy, and user acceptance [6], [8].

B. Biomedical Sensing Principles

Biomedical sensors are the foundational components of any intelligent monitoring system. Heart rate and blood-oxygen saturation are commonly obtained through photoplethysmography (PPG), an optical technique that measures changes in blood volume by illuminating a thin region of tissue—typically the fingertip—with red and infrared light and recording the fraction that is reflected back to a photodetector [9]. Because oxygenated and deoxygenated hemoglobin absorb red (≈ 660 nm) and infrared (≈ 880 – 940 nm) light differently, the ratio of the pulsatile components at the two wavelengths yields an estimate of SpO_2 , while the periodicity of the signal yields heart rate [10]. The MAX30102 sensor integrates both LEDs, a photodiode, and an 18-bit sigma-delta converter in a single low-power module, and has been characterized as a viable core for low-cost pulse oximeters [11]. Body temperature is acquired without contact through the MLX90614 infrared thermometer, which infers surface temperature from emitted thermal radiation [12].

C. Wireless Communication and Edge Processing

Whereas many monitoring systems rely on Wi-Fi or Bluetooth to forward data to cloud platforms, GSM cellular communication provides wide-area coverage independent of local network infrastructure, which is decisive for deployments in homes and rural areas with limited connectivity [13]. Processing the decision logic directly on the microcontroller—an edge-computing approach—reduces latency and bandwidth requirements and allows the system to issue alerts even without continuous internet access, in line with the broader shift toward edge intelligence in healthcare [14], [15].

D. Security and Privacy of Medical Data

The continuous acquisition and wireless transmission of physiological information raises significant security and privacy concerns, since health data are highly sensitive and subject to regulatory protection [16]. Surveys of e-health systems emphasize that medical devices, networks, and edge/cloud layers each introduce distinct attack surfaces, and recommend lightweight encryption, authenticated transmission, and privacy-preserving processing as design priorities for resource-constrained wearable devices [16], [17]. These requirements frame the present prototype as an early-stage proposal whose transition to clinical use would require the explicit incorporation of secure communication and data-protection mechanisms.

E. Related Systems

Several recent systems inform the present design. Mamdiwar et al. review IoT-assisted wearable sensor systems for healthcare monitoring, implementing PPG and ECG sensing with Arduino Nano and ESP8266 microcontrollers, Bluetooth modules, and rechargeable batteries for the wireless transmission of biomedical information [18]. Salem et al. present a non-invasive acquisition system that uses a MAX30100 sensor for heart rate and SpO_2 , infrared temperature sensors, and NodeMCU ESP8266 modules

feeding a cloud platform for real-time monitoring [19]. De Fazio et al. describe innovative IoT and wearable sensing solutions for human biophysical parameters, integrating SpO_2 and ECG sensors, ARM microcontrollers, Raspberry Pi, GSM modules, and thermal sensors for automated preventive supervision [20]. A complementary national reference is the prototype reported by Vela Mosquera for IoT-based health-indicator monitoring [21]. Table I summarizes the comparison between these systems and the present proposal.

TABLE I. COMPARATIVE ANALYSIS OF SMART BIOMEDICAL TECHNOLOGIES

System	Key components	Communication	Main function
Mamdiwar et al. [18]	PPG/ECG sensors, Arduino Nano, ESP8266, Bluetooth	Bluetooth	Wearable vital-signs supervision
Salem et al. [19]	MAX30100, IR temperature, NodeMCU ESP8266	Wi-Fi / cloud	Real-time biomedical acquisition
De Fazio et al. [20]	SpO_2 /ECG sensors, ARM, Raspberry Pi, GSM, thermal	GSM / Wi-Fi	Automated preventive supervision
This work	ESP32, MAX30102, MLX90614, SIM800L, LCD 16x2	GSM (SMS)	Vital-signs supervision with automatic alerts

As shown in Table I, the analyzed projects implement IoT technologies, biomedical sensors, and wireless communication to supervise vital signs in real time. The present proposal differs in that it combines GSM-based automatic alerting with biomedical automation as an accessible alternative oriented toward preventive monitoring and the autonomous generation of alerts when physiological anomalies are detected, without depending on permanent internet connectivity.

III. SYSTEM DESIGN AND IMPLEMENTATION

The proposed system shares core characteristics with the reviewed works—all focus on biomedical monitoring through IoT and intelligent sensors using programmable electronic devices that acquire and process physiological information in real time. The distinguishing feature of this proposal is the integration of GSM communication for the automatic transmission of alerts related to heart rate, oxygenation, and body temperature, implemented with low-cost components to maximize accessibility and adaptability.

A. Hardware Architecture

The prototype is built around an ESP32 microcontroller, selected for its dual-core processor, integrated wireless capabilities, and low cost. The MAX30102 sensor, placed in a finger clip, acquires heart rate and SpO_2 via reflective photoplethysmography, while the MLX90614 infrared thermometer measures body temperature without contact. A SIM800L module provides quad-band GSM connectivity for SMS alerting. A 16x2 I²C LCD displays current readings locally, an LM2596 DC-DC converter regulates the supply voltage, and a rechargeable battery enables portable operation. The components are housed in a compact enclosure, as shown in Fig. 1.

The ESP32 was preferred over single-core alternatives such as the ESP8266 and over Arduino-class boards because its dual-core Xtensa LX6 processor, operating at up to 240

MHz with 520 KB of SRAM, allows one core to manage the communication stack while the other executes the acquisition and decision logic without interruption [22]. This is advantageous in a continuous-monitoring application, where sampling, signal validation, local display, and alert dispatch must proceed concurrently. The microcontroller also integrates multiple I²C and UART interfaces—used here to connect the MAX30102 and MLX90614 sensors over I²C and the SIM800L module over UART—as well as hardware support for secure boot and flash encryption, which provides a foundation for the data-protection measures discussed in Section II-D [22]. Its low unit cost and broad open-source ecosystem further align with the accessibility objective of the proposal.

TABLE II. MAIN HARDWARE COMPONENTS OF THE PROTOTYPE

Component	Role	Interface / notes
ESP32	Main controller, acquisition and decision logic	Dual-core, Wi-Fi/BLE, I ² C/UART
MAX30102	Heart rate and SpO ₂ sensing	Reflective PPG, I ² C
MLX90614	Body-temperature sensing	Non-contact IR, I ² C
SIM800L	SMS alert transmission	Quad-band GSM, UART
LCD 16x2, LM2596, battery	Local display, power regulation, portability	I ² C, DC-DC, Li-ion



Fig. 1. Fig. 1. External and internal view of the physical prototype, showing the ESP32, SIM800L GSM module, LCD display, finger sensor, and the SMS alert received on a paired phone.

B. Firmware and Decision Logic

The ESP32 firmware periodically samples the three physiological variables, applies basic signal validation, and compares each reading against configurable safety thresholds. When a value falls outside the normal range—for example, heart rate above or below resting bounds, SpO₂ below a critical level, or elevated body temperature—the system classifies the state as an alert and triggers the GSM module to send an SMS. Each message details the specific values detected, a timestamp, and a recommendation to attend the nearest medical center. Processing the decision logic on the device follows an edge-computing approach that minimizes latency and allows alerts to be issued independently of internet availability [14].

C. Web Dashboard

In addition to local display and SMS alerting, a lightweight web dashboard centralizes the visualization of the acquired data. The interface presents the current heart rate, oxygenation, temperature, and system state, together with a historical record of measurements stored in a database, enabling caregivers or family members to interpret the biomedical information quickly and to detect anomalies through visual indicators and automatic alerts. Fig. 2 shows the dashboard, and Fig. 3 summarizes the projected impact of the prototype across six monitoring and well-being dimensions.



Fig. 2. Fig. 2. Web dashboard of the intelligent monitoring system, showing real-time heart rate, oxygenation, temperature, system state, and the historical record of measurements.

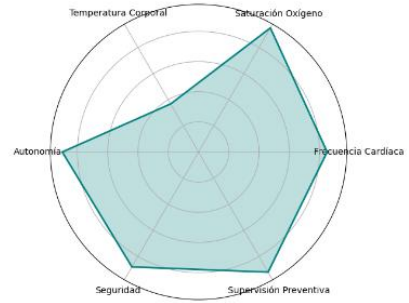


Fig. 3. Fig. 3. Radar diagram of the projected impact of the prototype across six dimensions: heart rate, oxygen saturation, body temperature, autonomy, safety, and preventive supervision.

IV. RESULTS AND DISCUSSION

Functional testing of the prototype confirmed the continuous acquisition and on-device processing of heart rate, SpO₂, and body temperature, the correct local display of readings, the centralized visualization through the web dashboard, and the reliable delivery of SMS alerts through the GSM network when simulated out-of-range values were introduced. The integration of biomedical sensors with GSM communication thus demonstrates the feasibility of an accessible preventive-monitoring solution that operates without permanent internet connectivity, consistent with the benefits reported for IoT-based vital-signs monitoring in the elderly [6], [8].

These results should nevertheless be interpreted as a proof of concept rather than a clinical validation. The thresholds were configured from general physiological references and were not personalized, and the measurements were not validated against certified medical-grade instruments; consumer PPG modules such as the MAX30102 require calibration and are sensitive to motion artifacts and peripheral perfusion, which can affect accuracy [11]. A rigorous

evaluation would require comparison against reference devices across a representative sample of older adults, the quantification of false-positive and false-negative alert rates, and an assessment of measurement stability over extended periods. Furthermore, as discussed in Section II-D, the current implementation does not yet incorporate the encryption and authentication mechanisms that the handling of sensitive health data demands [16], [17]; these are essential before any deployment beyond a controlled testbed.

V. PROJECTED IMPACT ON OLDER ADULTS

The prototype is designed to provide continuous, preventive supervision from the home. By detecting deviations in heart rate, oxygen saturation, and body temperature and issuing detailed alerts that recommend timely medical attention, the system aims to strengthen user safety, reduce the risk of unattended medical emergencies, and improve the responsiveness of caregivers and family members. The radar diagram in Fig. 3 illustrates the expected balance of benefits across physiological monitoring and well-being dimensions.

Equally important, the proposal seeks to increase the autonomy of older adults, allowing them to maintain an independent lifestyle without compromising health supervision. The ease of use, the accessibility of the components, and the possibility of remote oversight through the web dashboard and GSM alerts position the system as an integral solution that can contribute to quality of life and to the adoption of digital tools in the care of older people. These projected benefits remain hypotheses to be confirmed through field studies with end users and caregivers, including usability evaluation and acceptance analysis, which the literature identifies as decisive for the sustained adoption of wearable monitoring among older adults [8].

VI. CHALLENGES AND LIMITATIONS

Beyond the proof-of-concept results, several challenges must be addressed before a system of this kind can move toward real-world deployment. The first concerns measurement reliability. Optical PPG modules such as the MAX30102 are sensitive to motion artifacts, ambient light, skin tone, and peripheral perfusion, all of which are especially relevant in older adults, who may present reduced perfusion or involuntary movement; without calibration against certified instruments and artifact-rejection processing, readings can deviate from clinical-grade references [10], [11]. A robust system should therefore incorporate signal-quality indices and discard or flag low-confidence measurements rather than alerting on them.

A second challenge is the calibration of alert thresholds. Fixed, population-level limits inevitably trade sensitivity against specificity: thresholds that are too tight generate false alarms that erode user trust, while loose thresholds risk missing genuine deterioration. Personalization based on each user's baseline, and the use of trend analysis rather than single-sample comparisons, are promising directions for reducing both error types. A third challenge is energy autonomy: continuous sensing, GSM transmission, and local display impose a non-trivial power budget, so duty-cycling the radio, exploiting the ESP32 low-power modes, and dimensioning the battery for realistic operating times are practical requirements for a wearable form factor [22].

Finally, the handling of sensitive medical data introduces security, privacy, and regulatory obligations. As discussed in Section II-D, the current prototype transmits plain SMS messages and does not yet implement encryption, device authentication, or access control; integrating lightweight cryptographic mechanisms appropriate for resource-constrained devices, together with explicit consent and data-governance procedures, is a prerequisite for any clinical or community deployment [16], [17]. These limitations do not diminish the feasibility demonstrated by the prototype, but they delimit a concrete agenda for transforming it into a dependable digital-health tool.

VII. CONCLUSIONS

The automation of vital-signs monitoring through smart sensors and GSM communication constitutes an innovative and accessible technological solution for the continuous supervision of older adults' health. The development of the prototype showed that the integration of IoT technologies, biomedical sensors, and a microcontroller makes it possible to acquire, process, and transmit physiological information in real time, providing constant tracking of critical parameters such as heart rate, oxygen saturation, and body temperature.

The system incorporates automatic alerts delivered through GSM communication that detail the specific values detected and recommend attending the nearest medical center when an anomaly occurs. This functionality strengthens the preventive response, reduces the risks associated with delays in detecting health problems, and provides greater peace of mind to older adults and their caregivers, while the edge-based decision logic preserves operation under limited connectivity.

The prototype contributes to increasing the autonomy and safety of users and facilitates remote supervision by family members or health personnel; its accessibility, ease of use, and adaptability to domestic and community environments consolidate it as an effective tool for preventive health monitoring. At the same time, the analysis identified clear boundary conditions: the need for clinical validation against certified instruments, the personalization of thresholds, and the incorporation of security and privacy mechanisms for the protection of sensitive medical data.

Future work will therefore focus on a quantitative validation campaign against medical-grade references, the implementation of authenticated and encrypted communication, the personalization of alert thresholds, and a usability study with older adults and caregivers. Overall, the implementation of intelligent technologies in biomedical monitoring reflects a relevant trend in the digital transformation of the health sector, promoting preventive systems and digital-health solutions that can positively impact the quality of life of older adults and strengthen remote care for this vulnerable population.

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